Free pythagorean theorem worksheet worksheet 28

Maths | 10th grade | Mixed Operations

1. If 3x - 5y = 12 and 4x + 2y = 10, what is the value of x?

A) 1
B) 2
C) 3
D) 4

2. Simplify the expression: (2x^2 - 3x + 4) - (x^2 + 2x - 3).

A) x^2 - 5x + 7
B) x^2 - x + 1
C) 3x^2 - 5x + 7
D) x^2 - x + 7

3. What is the solution to the inequality: 5 - 2x > 3x + 10?

A) x < -1
B) x > -1
C) x < 3
D) x > 3

4. Evaluate the expression when a = -2 and b = 3: 4a^3 + 2b^2 - ab.

A) -22
B) -42
C) -36
D) -14

5. If the function f(x) = 2x^2 - 3x + 1 is transformed by shifting 2 units up, what is the new function?

A) 2x^2 - 3x + 3
B) 2x^2 - 3x + 5
C) 2x^2 - 3x - 1
D) 2x^2 - 3x + 1

6. Solve for x: 9^(x+1) = 27^(x-1).

A) 0
B) 1
C) 2
D) 3

7. Which of the following is the derivative of f(x) = 5x^3 - 2x + 4?

A) 15x^2 - 2
B) 15x^2 + 2
C) 5x^2 - 2
D) 15x^2

8. Find the least common multiple (LCM) of 12 and 18.

A) 36

B) 54
C) 72
D) 108

9. Solve for y in terms of x: 3^x = 81^(y+1).

A) y = (x-4)/8
B) y = 4x - 1
C) y = x/4 - 1
D) y = x/8 + 1

10. If the sum of a number and its reciprocal is 5/2, what is the number?

A) 2
B) 1/2
C) 3/2
D) 5/2

Answer Key

1. B
Solve the system of equations using substitution or elimination to find x = 2.

2. A
Subtract the second polynomial from the first: 2x^2 - x^2 - 3x - 2x + 4 + 3 = x^2 - 5x + 7.

3. A
Rearrange to -2x - 3x > 10 - 5, getting -5x > 5, hence x < -1.

4. C
Substitute a = -2 and b = 3 into the expression: 4(-2)^3 + 2(3)^2 - (-2)(3) = -32 + 18 + 6 = -36.

5. A
Adding 2 to the original function shifts it up: f(x) = 2x^2 - 3x + 1 + 2 = 2x^2 - 3x + 3.

6. B
Convert 9 to 3^2 and 27 to 3^3, then equate exponents: 2(x+1) = 3(x-1), solving gives x = 1.

7. A
Differentiate the function: f'(x) = 3*5x^2 - 2 = 15x^2 - 2.

8. A
The LCM is the smallest number divisible by both 12 and 18, which is 36.

9. A
Convert 81 to 3^4: 3^x = (3^4)^(y+1), so x = 4(y+1) gives x = 4y + 4, thus y = (x-4)/8.

10. C
Let the number be x. Then x + 1/x = 5/2. Solving gives x^2 - (5/2)x + 1 = 0, and x = 3/2 by the quadratic formula.